METRIC

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DETAIL SPECIFICATION

DIBUTYLTIN DILAURATE

This specification is approved for use by the Naval Surface Warfare Center Indian Head Division (NSWC IHD) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of dibutyltin dilaurate.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-STD-1916 - DoD Preferred Methods for Acceptance of Product

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D891 - Standard Test Methods for Specific Gravity, Apparent, of Liquid Industrial Chemicals

ASTM D1544 - Standard Test Method for Color of Transparent Liquids (Gardner Color Scale)

(Copies of these documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Product characteristics. The dibutyltin dilaurate shall meet the requirements of table I when tested in accordance with 4.5.

TABLE I. Chemical and physical properties.

Property	Requirement		Test Procedure
	Minimum	Maximum	
Tin content, %	17.7	19.5	4.5.1
Color, Gardner	--	4	4.5.2
Acid number	130	180	4.5.3
Specific gravity at 25 °C	1.045	1.054	4.5.4

3.3 Workmanship. The material shall be uniform in quality and free from foreign materials that could adversely affect its use when examined as specified in 4.5.5.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows (table II lists the applicable tests required for each particular level of inspection):

a. First article inspection (see 4.3).

b. Conformance inspection (see 4.4).

TABLE II. Applicable tests for inspection. 1/

Test	First Article Inspection	Conformance Inspection	Requirement
Tin content (4.5.1)	1	1	3.2
Color (4.5.2)	1	1	3.2
Acid number (4.5.3)	1	1	3.2
Specific gravity (4.5.4)	1	1	3.2
Workmanship (4.5.5)	1	1	3.3

NOTE:

1/ The numbers indicated are the number of unique lots (see 4.4.1.1) to be produced for the applicable testing required.

4.2 Test conditions. Unless otherwise specified in 4.5, test conditions shall be room ambient conditions.

4.3 First article inspection. When required (see 3.1), the first article inspection shall include the tests specified in table II. The sample size shall be as specified (see 6.2). Failure of the first article sample to pass this inspection shall be cause for rejection of the first article sample.

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4.4 Conformance inspection. Conformance inspection shall include the tests specified in table II.

4.4.1 Sampling.

4.4.1.1 Lot. For the purpose of sampling, a lot shall consist of material of a single form manufactured under one set of conditions from one production batch and offered for delivery at one time.

4.4.1.2 Sampling for workmanship. A random sample of filled containers shall be selected from each lot in accordance with MIL-STD-1916, verification level II.

4.4.1.3 Sampling for tests. A representative sample of sufficient material to form the required tests shall be taken from each lot.

4.5 Test procedures. When tested as specified in 4.5.1 through 4.5.5, the dibutyltin dilaurate shall meet the requirements of 3.2 and 3.3. Unless otherwise specified herein, all chemicals and reagents used in test methods shall be reagent grade. If the sample fails to pass one or more tests, the lot represented shall be rejected.

4.5.1 Tin content. Either the titrimetric method of 4.5.1.1 or the alternate instrumental method of 4.5.1.2 shall be used to determine the tin content.

4.5.1.1 Titrimetric method. The equipment, reagents, and procedure to determine the tin content using the titrimetric method shall be as follows.

4.5.1.1.1 Equipment and reagents. The equipment and reagents shall be as follows:

a. Erlenmeyer flasks, 500 milliliters (mL)

b. Hot plate

c. Meeker burner

d. Concentrated sulfuric acid, chemically pure reagent

e. Fuming nitric acid, chemically pure reagent

f. Hydrochloric acid, 2:1 (this acid should be boiled to remove air and placed in a stoppered bottle for future analyses)

g. Distilled water, boiled to remove all air

h. Standard 0.1N potassium iodate (prepared from water previously boiled)

i. Potassium iodide, reagent 10 percent solution (prepared from water previously boiled)

j. Starch solution (prepared from previously boiled water)

k. Iron powder, 99 percent iron hydrogen reduced

1. Nickel coils. Roll into a loose coil a 152.4- by 76.2- by 0.381-millimeter (6- by 3- by 0.015-inch) piece of nickel sheet. Before using the first time, clean in petroleum ether and then place in hydrochloric acid (2:1) and boil for about 5 minutes. Remove and rinse with water.

m. Perchloric acid (optional but preferred)

4.5.1.1.2 Procedure. The procedure shall be as follows.

4.5.1.1.2.1 Preparation of sample.

a. Weigh a 0.4- to 0.5-gram sample accurately to the nearest milligram (mg) via a 1-mL syringe into a 500-mL Erlenmeyer flask. Add 50 mL of fuming nitric acid. Swirl.

b. Add 20 mL of sulfuric acid. Swirl and add a few glass beads and 1 mL of perchloric acid.

c. Place on hot plate set at high heat; allow to boil until dense white fumes of sulfur trioxide appear. Fume for 10 minutes.

d. Using crucible tongs, swirl the flask over a Meeker burner and fume for 1 to 2 minutes. The solution should be colorless at this point. If not, add fuming nitric acid dropwise and bring to dense white fumes on hot plate. Repeat Meeker burner step.

e. Cool; wash down the sides of the flask with 30 mL of water. Place on hot plate and evaporate the solution to heavy fumes of sulfur trioxide.

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f. Cool and repeat step “e”.

g. Cool, add 80 mL of water, 100 mL of 2:1 hydrochloric acid, and 100 mL of water in that order.

4.5.1.1.2.2 Reduction and titration of tin.

a. To the prepared solution, add a nickel coil and 5 grams of iron powder. Swirl.

b. Wash down the sides of the flask with water.

c. Boil gently for about 20 minutes.

d. Cool. Add a few pieces of dry ice.

e. Add 5 mL of starch solution and 5 mL of 10 percent potassium iodide.

f. Titrate with potassium iodate to the starch-iodine endpoint.

g. Calculate the percentage of tin as follows:

$$\% \text{ Tin} = \frac{\text{Vol.} \times \text{N} \times 5.89}{\text{Wt. of sample}}$$

Where:

Vol. = mL of potassium iodate used

N = normality of potassium iodate

4.5.1.2 Instrumental method, atomic absorption. The equipment, materials, and procedure to determine the tin content using the instrumental method, atomic absorption, shall be as follows.

4.5.1.2.1 Equipment. The equipment shall be an atomic absorption spectrophotometer equipped with an acetylene and nitrous oxide gas system or any equivalent atomic absorption spectrophotometer.

4.5.1.2.2 Materials. The materials shall be American Chemical Society (ACS) grade xylenes, dibutyltin bis(2-ethylhexanoate), and an atomic absorption lamp for tin.

4.5.1.2.3 Procedure. The method shall follow the instructions prescribed by the manufacturer of the atomic absorption spectrophotometer for “tin in organometallic salts.” The sample weight shall be 0.6 gram of dibutyltin dilaurate weighed accurately to 0.1 mg. The final sample dilution shall represent 0.600 gram per 1000 mL. The wavelength shall be 286.3 nanometers. Standards and sample dilution may be altered to better suit capabilities of a particular instrument with absorbency of the solution falling between 0.2 and 1.0.

4.5.2 Gardner color. Gardner color shall be determined in accordance with ASTM D1544, except that the sample shall only be compared to color 4. The sample shall be recorded as a “pass” if the sample is lighter than or equal to color 4 and free of foreign material.

4.5.3 Acid number. The following procedure shall be performed to determine acid number.

4.5.3.1 Procedure. Weigh a 4- to 5-gram sample to the nearest mg on an analytical balance. Dissolve in 250 mL of isopropanol and 50 mL of water. (If a haze develops, add an additional 25 mL of isopropanol.) Titrate to a phenolphthalein endpoint using 1.0N potassium hydroxide (KOH) (pink color for at least 30 seconds). Calculate acid number as follows:

$$\text{Acid number} = \frac{\text{mL KOH} \times \text{normality} \times 56.1}{\text{Wt. of sample}}$$

4.5.4 Specific gravity. Specific gravity shall be determined in accordance with ASTM D891, test method B.

4.5.5 Workmanship. The samples taken in accordance with 4.4.1.2 shall be visually examined for contamination with extraneous material and uniformity using daylight illumination.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The dibutyltin dilaurate covered by this specification is intended for use in the manufacture of liner material for torpedo warheads and plastic-bonded explosives. Existing stock of dibutyltin dilaurate is acceptable for use until depleted.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Quantity desired.

c. When first article inspection is required (see 3.1 and 6.3).

d. First article sample size, if required (see 4.3 and 6.3).

e. Packaging requirements (see 5.1).

f. Safety data sheet (SDS) requirements (see 6.4).

6.3 First article. When first article inspection is required, the contracting officer will provide specific guidance to offerors whether the sample(s) will be a preproduction sample, a first article sample, a first production sample, a sample selected from the first production items, or a standard production sample(s) from the contractor's current inventory (see 3.1), and the size and number of samples to be tested as specified (see 4.3). The contracting officer will also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first article samples. When the dibutyltin dilaurate is used as a constituent of energetic materials required to comply with AOP-7 and STANAG 4170, the contracting officer will contact the Technical Authority (see 6.5) to determine the number and size of the samples and an acceptable laboratory for evaluation. Invitations for bids will provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product that has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation. Generally, a first article inspection, or part of a first article inspection, will be waived only if the following requirements have been met:

a. The manufacturer has made no changes to their process, procedures, or ingredient manufacturers since last qualified.

b. There has been no interruption in the manufacturer's production of dibutyltin dilaurate exceeding 12 months at the time of contract award.

6.4 SDSs. When required (see 6.2), contracting officers will identify those activities requiring copies of completed SDSs prepared in accordance with FED-STD-313. In order to obtain the SDS, FAR clause 52.223-3 must be in the contract.

6.5 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

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6.6 Subject term (key word) listing.

Catalyst

Energetic ingredient

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Navy – OS

Preparing activity:
Navy – OS
(Project 6810-2024-012)

Review activity:
DLA – GS

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